Supervisor Support for Recovery

Clarifying the Construct and Its Impact on Employee Recovery Experiences

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Why study supervisor support for recovery?

- **Recovery** is a relaxing process that reduces strain caused by work stressors; it sustains well-being, health, and performance.
- High demands and blurred work–nonwork boundaries make sufficient recovery increasingly difficult.
- Supervisors assign tasks, set deadlines, and communicate with employees, so they convey norms about nonwork time.
- Few studies focus on the role supervisors play in employees recovery experience, especially their ability to offer nonwork support help.

Research question

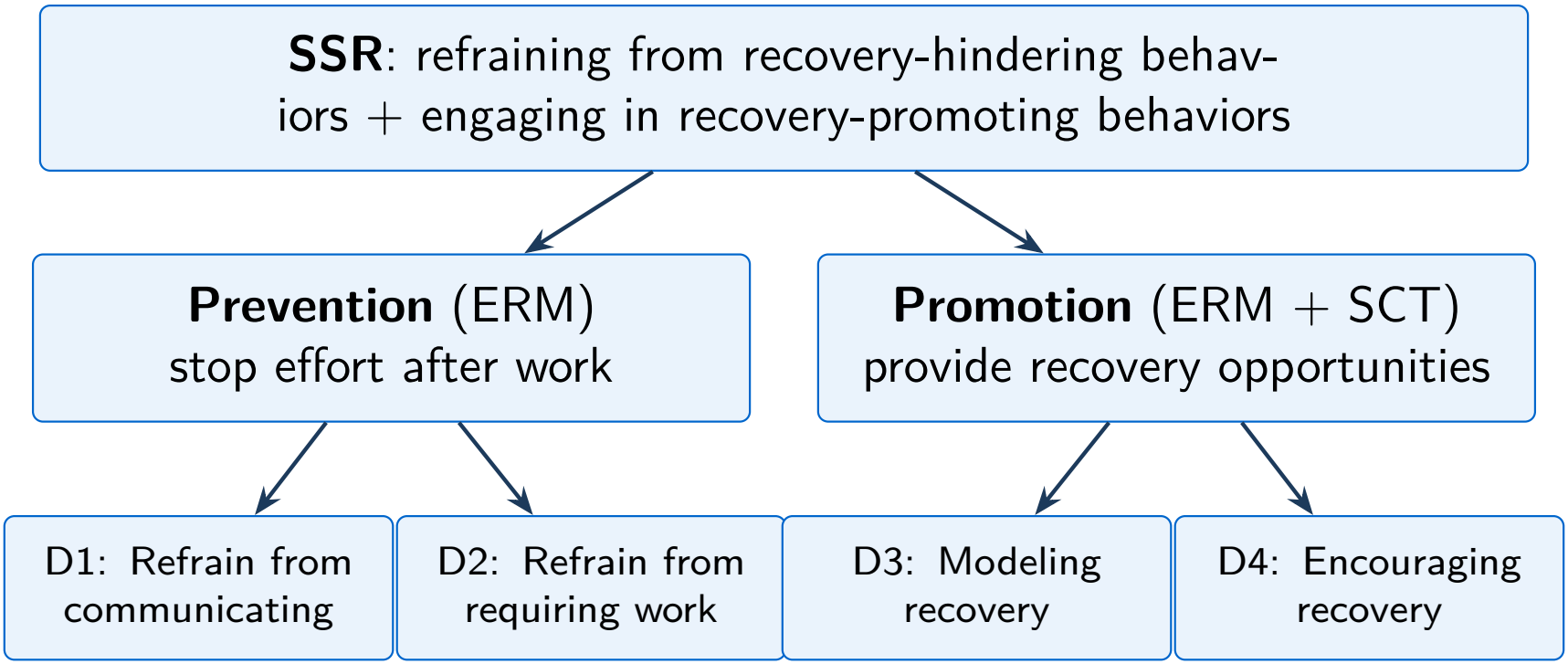
Which supervisor behaviors support employee recovery, how can they be measured, and do they explain recovery experiences beyond existing supervisor constructs?

From a narrow to an expanded construct

	Bennett et al. (2016)	Zhu et al. (2025)
Definition	Extent to which supervisors support recovery experiences at home	Tendency to refrain from recovery-hindering behaviors *and* engage in recovery-promoting behaviors
Focus	Expectation to work during nonwork time (imposing demands)	Prevention- *and* promotion-based supervisor behaviors
Structure	One dimension, 6 items, agree–disagree	Four dimensions, 20 items, frequency scale

Contribution: a multidimensional, behavior-based construct and a validated scale.

Theoretical foundations



- **ERM:** recovery occurs when effort expenditure from job demands redueces.
- **SCT:** behavior is learned through *modeling* (vicarious learning) and *encouragement* (verbal persuasion).

The four dimensions of SSR

Dimension	Direction	Definition and sample item	Study 1
D1: Refraining from communicating about work	Prevention	No texting, e-mailing, or calling about work during nonwork time. *"My supervisor only contacts me about work during working hours."*	57.4%
D2: Refraining from requiring work	Prevention	No task assignment or scheduling that compels work after hours. *"My supervisor makes last-minute requests that require me to work during nonwork time." (R)*	70.4%
D3: Modeling recovery	Promotion (modeling)	Setting an example of engaging in recovery. *"My supervisor uses time outside of work to wind down."*	11.1%
D4: Encouraging recovery	Promotion (persuasion)	Actively encouraging employees to get sufficient recovery. *"My supervisor encourages me to take time off."*	47.2%

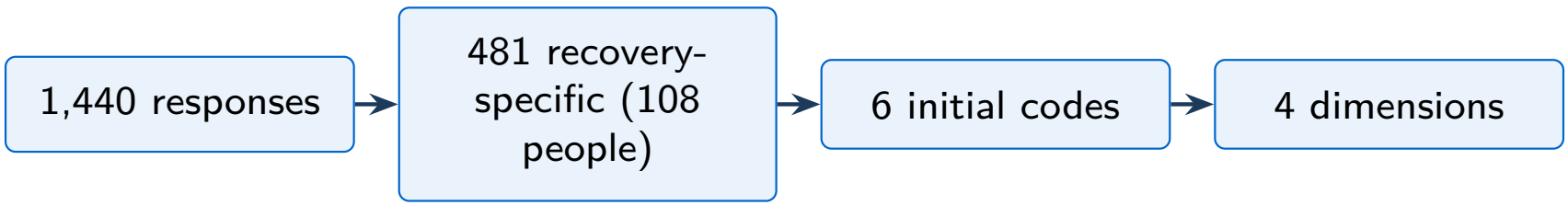
Research design: three phases, six studies

Phase	Study and sample	Purpose
1. Conceptual-ization	S1: MTurk, $N = 170$	Qualitative derivation of dimensions
	S2: 9 experts; 29 naive raters	Item writing, content validity
2. Psychometrics	S3a: MTurk, $N = 279$	Item analysis, EFA
	S3b: Prolific, $N = 451$	CFA, reliability
3. Validity	S4: U.S. employees, $N = 183$	Cross-sectional
	S5: Chinese dyads, $N = 350$	Multi-source
	S6: U.S. PhD students, $N = 570$	Time-separated (2-week lag)

Logic: each phase uses a new sample; Phase 3 varies rater, design, culture, and population.

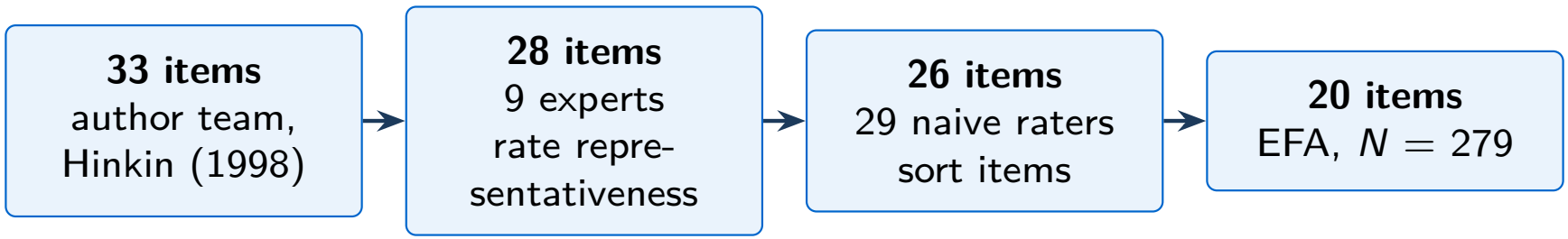
Study 1: qualitative derivation of the dimensions

- Full-time employees (MTurk, March 2020) listed supervisor behaviors that make it *harder* or *easier* to recover during nonwork time; Batch 2 added critical-incident follow-ups.
- **Coding:** inductive open coding → deductive grouping using recovery literature and SCT → re-sorting responses into dimensions.



- Inter-rater agreement: 93% for inclusion and 91% for dimension sorting (100 random responses each).
- 72.6% of responses described *hindering* behaviors; this may reflect frequency or negativity bias, which motivates a frequency-based behavioral scale.

Study 2 and Study 3a: building the item pool



- **Content validity** (Colquitt et al., 2019): definitional correspondence $p_{sa} = .66$–$.97$ ($M = .82$); distinctiveness $c_{sv} = .31$–$.93$ ($M = .63$). Two weak items dropped.
- **EFA:** maximum likelihood, oblique rotation; parallel analysis, scree plot, and eigenvalues all indicate four factors.
- Three items dropped for loadings $< .30$ or cross-loadings, and three for redundancy; final loadings .36–.91.
- **Response format:** 1 = *Never* to 5 = *Very frequently*, to capture behavior rather than evaluation.

Study 3b: confirmatory structure and reliability

- **What they did:** A new sample of 451 workers completed the 20-item questionnaire.
- **What they tested:** Do employees' answers support four related but distinct types of supervisor behavior?
  - Avoiding after-hours work communication
  - Avoiding demands for after-hours work
  - Setting an example of taking time to recover
  - Encouraging employees to rest and recover
- **Main finding:** Keeping these four types separate explained the response patterns better than combining them into fewer categories.
- **Consistency:** Questions within each type worked well together, supporting reliable measurement.
- **Meaning:** The questionnaire captures different ways supervisors support recovery. This study tests the measure; later studies examine its links with employee recovery.

Studies 4–6: validation designs

	Study 4	Study 5	Study 6
Sample	U.S. full-time white-collar (Prolific)	Chinese supervisor–subordinate dyads (Idiaoyan)	U.S. PhD students (e-mail)
N	183	350 dyads	570 (T1); 462 (T2)
Design	Cross-sectional	Multi-source	Time-separated, 2 weeks
SSR rated by	Employee	Supervisor (self)	Student (about advisor)
Recovery rated by	Employee	Employee	Student at T2
Comparison constructs	PSS, LMX, FSSB, HoL	PSS, LMX, FSSB	PSS, LMX, FSSB-SF
SSR CFA fit	CFI .93, RMSEA .08	CFI .92, RMSEA .06	CFI .93, RMSEA .07

- **Outcomes:** Recovery Experience Questionnaire; psychological detachment (PD), relaxation, control, mastery (4 items each).
- **Control:** quantitative workload.

Hypotheses and analytic strategy

Hypothesis	Test	Decision rule
H1 Discriminant: SSR is distinct from PSS, LMX, HoL, FSSB	Nested CFAs: four SSR factors + comparison construct vs. a model merging *one* SSR dimension with that construct	Significant $\Delta\chi^2$ for every merge
H2 Criterion: SSR relates positively to the four recovery experiences	Bivariate correlations per study, then a fixed-effect internal meta-analysis ($k = 3$)	95% CI of mean r excludes 0
H3 Incremental: SSR predicts beyond similar constructs	Hierarchical regression: Step 1 workload; Step 2 PSS, LMX, FSSB (+HoL in S4); Step 3 *one* SSR dimension at a time	Significant ΔR^2 for at least one dimension

H1: discriminant validity

Merged with an SSR dimension	Study 4	Study 5	Study 6
PSS	288–716	164–433	793–1354
LMX	224–764	76–546	708–1231
HoL	315–811	—	—
FSSB	280–481[a]	88–670	385–763

Cells show the range of $\Delta\chi^2$ across the four merged models ($\Delta df = 4$); all $p < .001$. [a]$\Delta df = 7$; each SSR dimension merged with its most correlated FSSB dimension.

- **H1 supported in all three studies.**
- Smallest distinctions: D3 and D4 versus LMX and FSSB in Study 5 ($\Delta\chi^2 = 76$–110).
- encouraging recovery correlates highly with general support in Study 4 (PSS $r = .72$; HoL $r = .72$).

H2: criterion-related validity

Mean r [95% CI]	PD	Relaxation	Control	Mastery
D1 Communicating	**.34** [.28,.40]	.22 [.16,.28]	.29 [.23,.35]	.09 [.02,.15]
D2 Requiring work	.29 [.23,.35]	.28 [.22,.34]	.38 [.32,.43]	.09 [.03,.16]
D3 Modeling	.24 [.18,.30]	.23 [.17,.29]	.29 [.23,.35]	.15 [.09,.21]
D4 Encouraging	.24 [.18,.30]	.32 [.26,.38]	**.40** [.35,.45]	.28 [.22,.34]

Fixed-effect internal meta-analysis of Studies 4–6 ($k = 3$).

- **H2 fully supported** at the meta-analytic level: all 16 CIs exclude zero.
- Prevention matters most for detachment; encouragement matters most for control.
- Mastery is weakest: only 5 of 12 study-level correlations are significant.

H3: incremental validity (ΔR^2 at Step 3)

Study	Dimension	PD	Relaxation	Control	Mastery
S4 (cross-sectional)	D1	.11**	.06**	.04**	.01
	D2	.08**	.04**	.06**	.01
	D3	.02*	.02	.04**	.00
	D4	.00	.03*	.00	.00
S5 (multi-source)	D1	.06**	.00	.03**	.00
	D2	.00	.00	.01	.00
	D3	.04**	.01*	.01**	.01
	D4	.00	.01	.03**	.03**
S6 (time-separated)	D1	.02**	.00	.01*	.00
	D2	.02**	.00	.04**	.00
	D3	.01	.00	.01*	.00
	D4	.01*	.00	.02**	.00

Controls: workload, PSS, LMX, FSSB (+HoL in S4). $^{*}p < .05$, $^{**}p < .01$.

- **H3 partially supported:** robust for PD and control; mastery only in Study 5.
- Increments are small ($\Delta R^2 \leq .11$), generally smaller with separated raters or time.

Contributions and implications

- **Recovery literature:** extends supervisor support from limiting demands to also *providing* recovery opportunities.
- **Theory:** brings SCT (modeling, encouragement) into recovery research alongside ERM.
- **Practice:** behavior-specific dimensions can guide supervisor training, a contextual alternative to individual-level interventions.
- **Authors' future research:** mediators (energy, self-efficacy), moderators of mastery, finer time intervals, organizational samples.

Methodological concerns

- **Causality:** All studies are observational. Study 6 measures recovery two weeks after SSR but does not control for initial recovery. It therefore tests later recovery levels, not improvement in recovery, and cannot establish that supervisor support causes better recovery.
- **Item wording and factor structure:** All D2 items describe recovery-hindering behaviors and are reverse-scored, whereas D1 items describe supportive behaviors. The separation between D1 and D2 may partly reflect wording differences rather than distinct content. Testing items with more balanced wording would strengthen the four-factor interpretation.
- **Scope of incremental validity:** SSR dimensions are added to the regression one at a time. These analyses show whether each dimension adds information beyond existing leadership measures, but do not establish its unique contribution beyond the other SSR dimensions or the additional variance explained by all four dimensions together.

Questions for discussion

1. Should SSR be measured from the employee's view, the supervisor's view, or both? What would disagreement between them mean?
2. Do the four dimensions of SSR affect employee recovery through different mechanisms, such as energy preservation, recovery self-efficacy, and disconnection from work? (Xu Mengyao)
3. Would employees really see "no messages after work" as support from the supervisor, or just as the normal state of things? (Yu Guo)

The central takeaway

Construct

Supervisor support for recovery has two directions (prevention and promotion) and four behavioral dimensions.

Measurement

A 20-item frequency scale with a four-factor structure replicated across four samples, two countries, and two rater perspectives.

Validity

SSR is distinct from general and family-supportive supervision and adds explanatory power, mainly for psychological detachment and control, but not reliably for mastery.